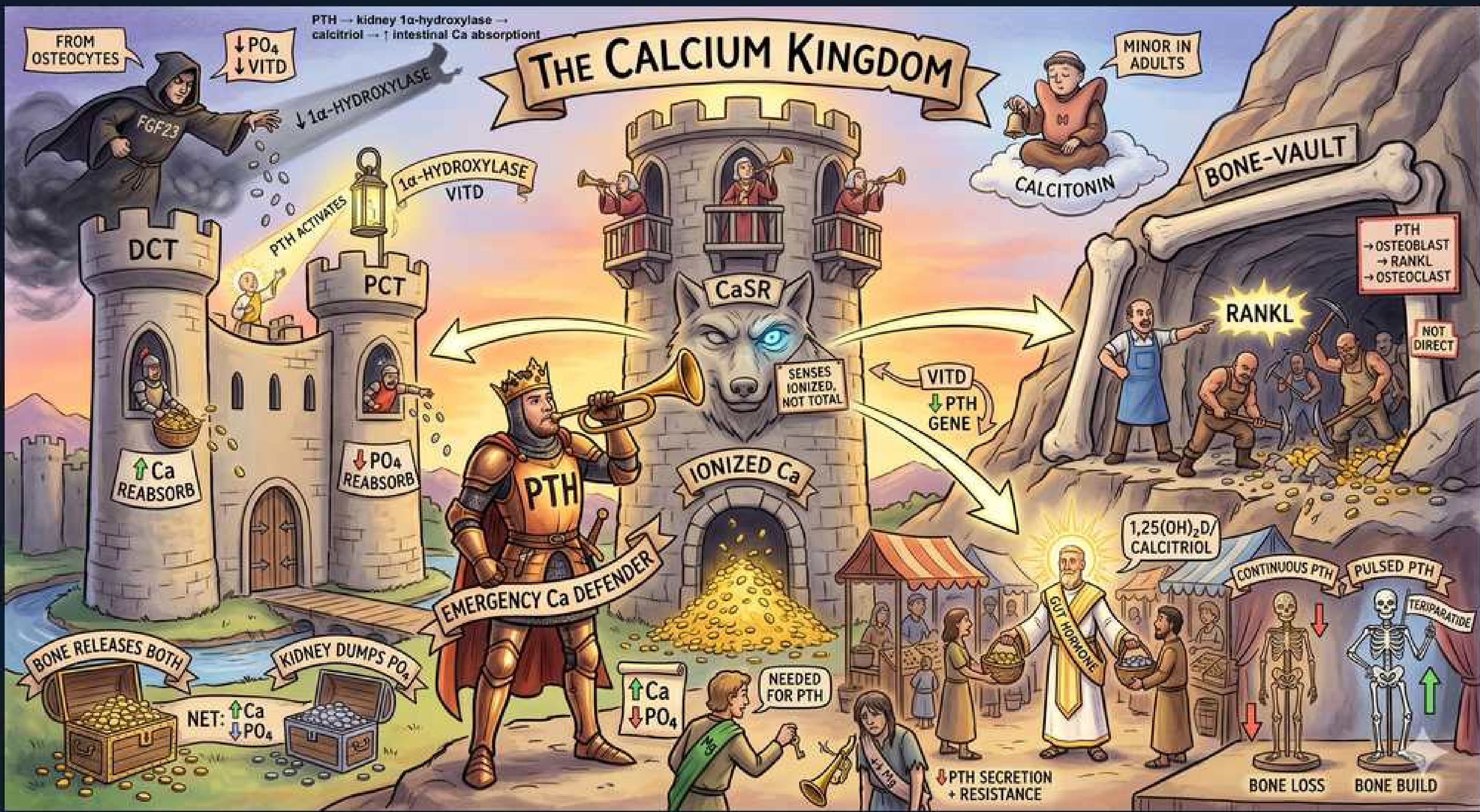
Calcium/PTH/VitD — The Calcium Kingdom



1. FGF23 from osteocytes

WHAT YOU SEE

A winged FGF23 figure flying out of bone with down-arrows for phosphate and vitamin D.

WHAT IT ENCODES

FGF23 is a phosphaturic hormone secreted by osteocytes/osteoblasts in response to high phosphate and high calcitriol. It does two things, both lowering: it promotes renal phosphate wasting (downregulates the proximal tubule Na-Pi cotransporter) AND it suppresses 1-alpha-hydroxylase, dropping calcitriol. Clinically, FGF23 excess underlies tumor-induced osteomalacia and X-linked hypophosphatemic rickets (low PO4, low/normal 1,25-D); in CKD, FGF23 rises early as a compensatory phosphaturic signal. Mnemonic: 'FGF23 = Phosphate Garbage Flush, vitamin D off.'

BOARD TRAP

A stem with low phosphate asks which hormone RAISES active vitamin D and you pick FGF23 — WRONG. FGF23 INHIBITS 1-alpha-hydroxylase and lowers calcitriol; only PTH raises calcitriol. The discriminator: FGF23 and PTH both waste phosphate, but they move calcitriol in OPPOSITE directions (FGF23 down, PTH up).

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2. PTH actions banner

WHAT YOU SEE

Top scroll listing PTH effects: kidney 1-hydroxylase, calcitriol, intestinal Ca absorption.

WHAT IT ENCODES

PTH's three net actions all RAISE serum calcium: (1) bone — stimulates resorption (via osteoblast RANKL), (2) kidney — increases distal Ca reabsorption and wastes phosphate proximally, and (3) gut — INDIRECTLY boosts calcium absorption by stimulating renal 1-alpha-hydroxylase to make calcitriol. Normal total serum calcium is ~8.5-10.5 mg/dL; correct measured total calcium by +0.8 mg/dL for each 1 g/dL the albumin falls below 4. Net effect of PTH: Ca up, PO4 down, calcitriol up.

BOARD TRAP

A question states PTH 'directly stimulates intestinal calcium absorption' — WRONG. PTH has no direct gut action; intestinal Ca uptake is mediated entirely by calcitriol (1,25-D), which PTH induces upstream. The discriminator: PTH acts directly on bone and kidney, but only indirectly (via vitamin D) on the gut.

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3. 1-alpha-hydroxylase lamp

WHAT YOU SEE

A glowing lantern labeled 1-alpha-hydroxylase converting to active vitamin D in the proximal tubule.

WHAT IT ENCODES

Renal 1-alpha-hydroxylase (CYP27B1) in the proximal convoluted tubule is the RATE-LIMITING and tightly regulated step converting 25-OH-D (calcidiol, the storage/marker form) to 1,25-(OH)2-D (calcitriol, the active form). It is stimulated by PTH and low phosphate, and inhibited by FGF23 and high calcitriol (feedback). Sarcoidosis/lymphoma express extrarenal 1-alpha-hydroxylase in macrophages, causing PTH-independent calcitriol-mediated hypercalcemia.

BOARD TRAP

A stem swaps the organs: 'the kidney makes 25-OH-D and the liver activates it' — WRONG. The LIVER makes 25-OH-D (storage form, best test of vitamin D status), and the KIDNEY makes the ACTIVE 1,25-D. The discriminator: 25 = liver (2-letter middle organ), 1 = kidney (final activation).

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4. DCT Ca reabsorb tower

WHAT YOU SEE

A castle tower labeled DCT with a defender pulling calcium back in.

WHAT IT ENCODES

PTH increases active calcium reabsorption in the distal convoluted tubule (via TRPV5 channels), conserving calcium and lowering urinary calcium. This is why primary hyperparathyroidism can paradoxically still cause hypercalciuria — the filtered calcium load from hypercalcemia overwhelms the enhanced reabsorption. Thiazide diuretics also act at the DCT to increase calcium reabsorption (useful in hypercalciuric stone formers; a cause of mild hypercalcemia).

BOARD TRAP

A stem claims PTH increases URINARY calcium excretion — WRONG at the tubular level. PTH intrinsically PROMOTES distal calcium reabsorption (lowers urine Ca); any hypercalciuria in hyperparathyroidism comes from the increased filtered load, not from a direct calciuric PTH effect. Loop diuretics (not PTH) waste calcium in the urine.

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5. PCT phosphate dumping

WHAT YOU SEE

Proximal tubule tower throwing phosphate (PO4) out into the moat.

WHAT IT ENCODES

PTH inhibits the proximal tubule sodium-phosphate cotransporter (NaPi-2a), producing phosphaturia and lowering serum phosphate. This is the renal handle behind the classic primary hyperparathyroidism pattern: high Ca with LOW phosphate. The urinary cAMP rises (PTH acts via Gs/cAMP), historically the basis of the Ellsworth-Howard test for pseudohypoparathyroidism.

BOARD TRAP

A stem expects you to think PTH raises BOTH calcium and phosphate — WRONG. PTH raises calcium but LOWERS phosphate (renal wasting); calcium and phosphate move in OPPOSITE directions with PTH. Contrast calcitriol, which raises BOTH. The discriminator is the phosphate direction: PTH down, calcitriol up.

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6. CaSR wolf sensor

WHAT YOU SEE

A wolf-head shield with glowing eyes labeled CaSR that senses ionized Ca, not total.

WHAT IT ENCODES

The calcium-sensing receptor (CaSR) is a Gq-coupled GPCR on parathyroid chief cells (and the renal thick ascending limb) that continuously samples IONIZED (free) calcium. When ionized Ca rises, CaSR activation SUPPRESSES PTH secretion; when it falls, PTH is released. It is the master setpoint of calcium homeostasis and the target of the calcimimetic cinacalcet (which lowers PTH in hyperparathyroidism).

BOARD TRAP

A question states the parathyroid senses TOTAL or albumin-bound calcium — WRONG. The CaSR detects only the IONIZED (free) fraction. This is why an acidemic or low-albumin patient can have a normal total calcium but a clinically relevant ionized calcium, and why PTH tracks ionized, not total, calcium.

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7. PTH king with trumpet

WHAT YOU SEE

A crowned king labeled PTH blowing a trumpet, called the Emergency Ca Defender.

WHAT IT ENCODES

PTH is the rapid, minute-to-minute defender of serum calcium, raising it through bone resorption, distal renal reabsorption, and (slower, indirect) calcitriol-mediated gut absorption. It is an 84-amino-acid peptide with a very short half-life (~2-4 min), enabling tight moment-to-moment control. Vitamin D/calcitriol provides the slower, longer-term regulation of calcium balance.

BOARD TRAP

A stem frames calcitriol as the FAST emergency responder to acute hypocalcemia — WRONG. PTH is the rapid responder (short half-life, immediate bone/kidney action); calcitriol is the slow, sustained regulator. The discriminator: acute drop in ionized Ca → PTH surges within minutes; vitamin D effects take hours to days.

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8. Ionized Ca treasure

WHAT YOU SEE

A pile of gold coins labeled Ionized Ca, marked 'needed for PTH'.

WHAT IT ENCODES

Ionized (free) calcium is the physiologically active fraction (~45-50% of total), the form sensed by the CaSR, and the determinant of neuromuscular excitability. Total calcium also includes albumin-bound (~40%) and complexed (~10%) fractions. Alkalosis increases calcium binding to albumin, dropping ionized Ca and causing tetany/Chvostek-Trousseau signs despite a normal total; acidosis does the reverse.

BOARD TRAP

A stem reports a low TOTAL calcium in a hypoalbuminemic patient and labels them truly hypocalcemic — WRONG without correcting. Use corrected Ca = measured + 0.8 x (4 - albumin); often the ionized calcium is normal. The discriminator: symptoms track IONIZED calcium, so check ionized (or corrected) calcium before treating.

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9. Calcitonin minor figure

WHAT YOU SEE

A small monk labeled Calcitonin tagged 'minor in adults'.

WHAT IT ENCODES

Calcitonin, from thyroid parafollicular C-cells, lowers calcium by inhibiting osteoclasts, but it plays a negligible role in normal adult calcium homeostasis. Its clinical value is as a TUMOR MARKER for medullary thyroid carcinoma (and screening in MEN2) and as a short-term, rapidly-acting (but tachyphylaxis-prone) adjunct in severe hypercalcemia while bisphosphonates take effect.

BOARD TRAP

A stem implies thyroidectomy causes hypercalcemia (loss of calcitonin) or that medullary thyroid cancer causes hypocalcemia (calcitonin excess) — BOTH WRONG. Calcitonin is physiologically minor in adults, so neither its loss nor its excess meaningfully disturbs serum calcium. It matters as an MTC marker, not as a calcium regulator.

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10. RANKL / osteoblast control

WHAT YOU SEE

Bone-vault scene: PTH directs osteoblasts which express RANKL to activate osteoclasts, marked 'NOT direct'.

WHAT IT ENCODES

Osteoclasts have NO PTH receptor. PTH binds OSTEOBLASTS (and osteocytes), which respond by upregulating RANKL and downregulating osteoprotegerin (OPG, the decoy receptor); RANKL then engages RANK on osteoclast precursors to drive maturation and bone resorption. Denosumab is a monoclonal anti-RANKL antibody that blocks this axis to treat osteoporosis and hypercalcemia of malignancy.

BOARD TRAP

A stem says PTH directly stimulates osteoclasts to resorb bone — WRONG. PTH acts on OSTEOBLASTS, which signal osteoclasts via RANKL/RANK; the effect on osteoclasts is indirect. The discriminator: the RANKL bridge is the only path PTH has to the osteoclast, which is why anti-RANKL (denosumab) potently blocks PTH-driven resorption.

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11. Calcitriol releasing Ca+PO4

WHAT YOU SEE

A robed figure labeled 1,25(OH)2D / Calcitriol holding up both Ca and PO4 up-arrows.

WHAT IT ENCODES

Calcitriol (1,25-(OH)2-D) raises BOTH serum calcium AND phosphate, mainly by increasing intestinal absorption of each (and mobilizing both from bone). This makes it the only major calcium hormone that raises phosphate. Vitamin D deficiency therefore causes low Ca and low PO4 with secondary high PTH (rickets/osteomalacia); calcitriol or its analogs are used in CKD and hypoparathyroidism.

BOARD TRAP

A stem assumes everything that raises calcium also lowers phosphate (like PTH) and applies that to calcitriol — WRONG. Calcitriol raises calcium AND phosphate together. The discriminator: PTH = Ca up / PO4 down; calcitriol = Ca up / PO4 up. Use phosphate direction to tell the two hypercalcemic hormones apart.

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12. Continuous vs pulsed PTH

WHAT YOU SEE

Two skeletons: continuous PTH causing bone loss, pulsed (intermittent) PTH building bone.

WHAT IT ENCODES

PTH effect on bone is dose/duration dependent. CHRONIC, continuous elevation (primary or secondary hyperparathyroidism) is catabolic, driving net resorption and osteitis fibrosa cystica. INTERMITTENT (once-daily) PTH exposure is paradoxically ANABOLIC, favoring osteoblastic bone formation — the basis of teriparatide (PTH 1-34) and abaloparatide for severe osteoporosis. Anabolic agents carry a black-box osteosarcoma warning and are typically limited to ~2 years.

BOARD TRAP

A stem treats all PTH as bone-destructive and excludes teriparatide as an osteoporosis therapy — WRONG. INTERMITTENT PTH analogs are bone-anabolic and indicated for high-fracture-risk osteoporosis. The discriminator: continuous PTH = bone loss; once-daily pulsed PTH = bone building.
